

DOKUMENTASI JARINGAN

Lab Simulasi Jaringan Warnet / Perusahaan

Domain: warnet.net

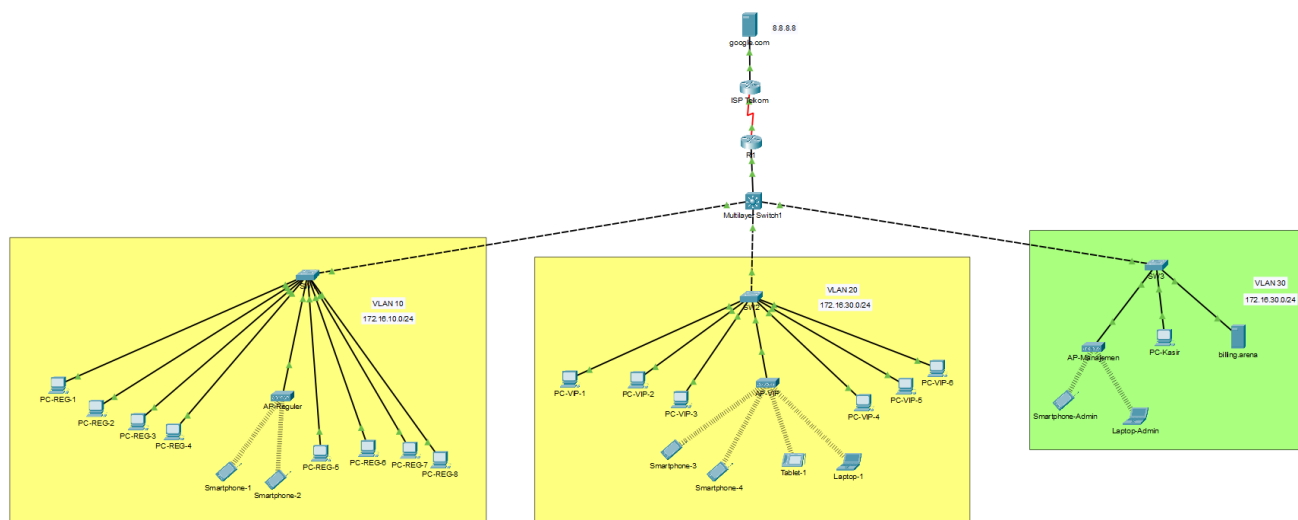
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Cisco Packet Tracer – Router, Multilayer Switch, VLAN, DHCP, ACL, NAT, SSH

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1. Ringkasan Topologi Jaringan

Jaringan lab ini mensimulasikan infrastruktur jaringan sebuah warnet/perusahaan kecil dengan koneksi ke internet melalui ISP, satu router utama sebagai gateway dan firewall dasar, satu multilayer switch sebagai backbone, serta tiga switch akses yang masing-masing melayani satu VLAN berbeda. Terdapat pula dua Access Point nirkabel untuk perangkat mobile pada VLAN Regular dan VLAN VIP, serta satu Access Point manajemen pada VLAN Manajemen.



Gambar 1. Topologi jaringan lab (Cisco Packet Tracer)

1.1 Alur Koneksi Utama

- Internet (server google.com, 8.8.8.8) — terhubung ke router ISP-Telkom melalui GigabitEthernet0/1 (8.8.8.1/24).
- ISP-Telkom — R1: koneksi WAN Serial0/0/0, subnet 202.152.100.0/30 (ISP-Telkom = .2, R1 = .1).
- R1 — MLS1: koneksi trunk 802.1Q melalui GigabitEthernet0/0, membawa VLAN 10, 20, dan 30 (router-on-a-stick).
- MLS1 — SW1/SW2/SW3: tiga link trunk (Gig1/0/1, Gig1/0/2, Gig1/0/3) menuju masing-masing switch akses.
- SW1 (VLAN 10 – Regular): melayani PC-REG-1 s.d. PC-REG-8 dan Access Point "AP-Reguler" (Smartphone-1, Smartphone-2).
- SW2 (VLAN 20 – VIP): melayani PC-VIP-1 s.d. PC-VIP-6 dan Access Point "AP-VIP" (Smartphone-3, Smartphone-4, Tablet-1, Laptop-1).
- SW3 (VLAN 30 – Manajemen): melayani PC-Kasir, server billing.arena, dan Access Point "AP-Manajemen" (Smartphone-Admin, Laptop-Admin).
- Manajemen jaringan (SSH) hanya mencakup perangkat internal warnet — R1, MLS1, SW1, SW2, SW3. ISP-Telkom berada di luar batas administratif warnet sehingga tidak termasuk dalam lingkup manajemen ini (lihat bagian 8).

2. Daftar Perangkat dan Peran

Hostname	Peran	Platform	IP Manajemen
ISP-Telkom	Router ISP (simulasi internet)	Cisco 2911	202.152.100.2/30, 8.8.8.1/24
R1	Router utama (gateway, DHCP, NAT, ACL, routing)	Cisco 2911	202.152.100.1/30, sub-interface VLAN

Hostname	Peran	Platform	IP Manajemen
MLS1	Multilayer Switch (backbone / trunk aggregator)	Catalyst L3 (16.3.2)	172.16.30.2/24 (VLAN30)
SW1	Switch akses VLAN 10 (Reguler)	Catalyst L2	172.16.30.3/24 (VLAN30)
SW2	Switch akses VLAN 20 (VIP)	Catalyst L2	172.16.30.4/24 (VLAN30)
SW3	Switch akses VLAN 30 (Manajemen)	Catalyst L2	172.16.30.5/24 (VLAN30)

Catatan: SW1, SW2, SW3, dan MLS1 tidak memiliki alamat IP pada VLAN 1 (interface Vlan1 dalam kondisi shutdown, tanpa IP). Manajemen keempat perangkat ini dilakukan melalui SVI VLAN 30, sehingga hanya dapat diakses dari jaringan Manajemen.

3. Skema VLAN dan Pengalamatan IP

3.1 Tabel VLAN

VLAN	Nama	Network	Gateway (R1)	Switch Akses
10	Reguler	172.16.10.0/24	172.16.10.1	SW1
20	VIP	172.16.20.0/24	172.16.20.1	SW2
30	Manajemen	172.16.30.0/24	172.16.30.1	SW3

3.2 Interface Router R1 (Router-on-a-Stick)

Interface	Encapsulation	IP Address	Keterangan
Gi0/0	-	no ip address	Interface fisik, trunk pembawa sub-interface
Gi0/0.10	dot1Q VLAN 10	172.16.10.1/24	Gateway VLAN Reguler, ACL "REGULER" (in)
Gi0/0.20	dot1Q VLAN 20	172.16.20.1/24	Gateway VLAN VIP, ACL "VIP" (in)
Gi0/0.30	dot1Q VLAN 30	172.16.30.1/24	Gateway VLAN Manajemen
Se0/0/0	-	202.152.100.1/30	Link WAN ke ISP-Telkom

3.3 Link WAN dan Internet

Segment	Network	Perangkat A	Perangkat B
WAN R1–ISP	202.152.100.0/30	R1 Se0/0/0 (.1)	ISP-Telkom Se0/0/0 (.2)
Simulasi Internet	8.8.8.0/24	ISP-Telkom (8.8.8.1) Gi0/1	Server google.com (8.8.8.8)

4. Konfigurasi Trunk dan Port Akses

4.1 Link Trunk (802.1Q)

Perangkat	Interface	Mode	Terhubung ke
MLS1	Gig1/0/24	Trunk	R1 Gi0/0 (router-on-a-stick)

Perangkat	Interface	Mode	Terhubung ke
MLS1	Gig1/0/1	Trunk	SW1 (VLAN 10)
MLS1	Gig1/0/2	Trunk	SW2 (VLAN 20)
MLS1	Gig1/0/3	Trunk	SW3 (VLAN 30)

4.2 Alokasi Port Akses per Switch

Switch	VLAN	Port Akses (switchport access vlan)	Port Tidak Terpakai
SW1	10	Fa0/2 – Fa0/10	Fa0/11–24, Gi0/1–0/2 → VLAN 999 (shutdown)
SW2	20	Fa0/2 – Fa0/8	Fa0/1, Fa0/9 – Fa0/24 (belum dikonfigurasi)
SW3	30	Fa0/2 – Fa0/4	Fa0/1, Fa0/5 – Fa0/24 (belum dikonfigurasi)

SW1 kini sudah menerapkan praktik keamanan port yang baik: seluruh port yang tidak digunakan (Fa0/11–Fa0/24, Gi0/1, Gi0/2) dipindahkan ke VLAN 999 (VLAN "black hole" tanpa fungsi routing) dan diberi status shutdown, sehingga tidak dapat langsung digunakan meski ada yang mencolokkan kabel secara fisik. Praktik yang sama sebaiknya diterapkan juga pada SW2 dan SW3 (lihat bagian Rekomendasi).

5. Konfigurasi DHCP (pada R1)

R1 bertindak sebagai DHCP server untuk ketiga VLAN. Sepuluh alamat pertama pada setiap subnet (.1 – .10) dikecualikan dari pool karena dipakai untuk gateway dan alamat manajemen perangkat.

Pool	Network	Default Gateway	DNS Server	Excluded Address
REGULER_POOL	172.16.10.0/24	172.16.10.1	8.8.8.8	172.16.10.1 – .10
VIP_POOL	172.16.20.0/24	172.16.20.1	8.8.8.8	172.16.20.1 – .10
MANAJEMEN_POOL	172.16.30.0/24	172.16.30.1	8.8.8.8	172.16.30.1 – .10

6. Access Control List (Segmentasi Antar VLAN)

R1 menerapkan ACL extended pada masing-masing sub-interface VLAN Reguler dan VIP untuk mencegah kedua VLAN tersebut mengakses jaringan Manajemen (VLAN 30), sementara akses ke jaringan lain (termasuk internet) tetap diizinkan.

Nama ACL	Diterapkan pada	Arah	Aturan
REGULER	Gi0/0.10	in	deny 172.16.10.0/24 → 172.16.30.0/24; permit ip any any
VIP	Gi0/0.20	in	deny 172.16.20.0/24 → 172.16.30.0/24; permit ip any any

Dengan konfigurasi ini, PC-REG-x dan PC-VIP-x/AP terkait tidak dapat menjangkau perangkat manajemen (PC-Kasir, billing.arena, SW1/SW2/SW3, MLS1), namun tetap dapat mengakses internet melalui NAT di R1.

7. Network Address Translation (NAT)

R1 melakukan NAT overload (PAT) agar seluruh host pada VLAN 10, 20, dan 30 dapat mengakses internet menggunakan satu alamat IP publik pada interface Serial0/0/0.

Komponen	Konfigurasi
ACL sumber NAT	NAT_ACL — permit 172.16.10.0/24, 172.16.20.0/24, 172.16.30.0/24
Perintah NAT	ip nat inside source list NAT_ACL interface Serial0/0/0 overload
Interface publik	Serial0/0/0 (202.152.100.1/30)
Default route R1	ip route 0.0.0.0 0.0.0.0 202.152.100.2 (menuju ISP-Telkom)

Perlu diverifikasi: pada hasil show running-config yang diberikan, perintah ip nat inside pada sub-interface VLAN dan ip nat outside pada Serial0/0/0 tidak terlihat eksplisit. Pastikan kedua perintah ini benar-benar diterapkan di perangkat agar NAT berfungsi.

8. Manajemen Perangkat dan Akses Remote

Manajemen jarak jauh (SSH) pada jaringan ini secara sengaja dibatasi hanya untuk perangkat internal milik warnet — yaitu R1, MLS1, dan SW1–SW3. ISP-Telkom merepresentasikan perangkat di luar kendali administratif warnet (milik/dikelola pihak ISP), sehingga tidak disertakan dalam lingkup manajemen SSH internal ini.

Perangkat	SSH v2	Enable Secret	User (hash) Secret	VTY Login	Access-class VTY
R1	Ya	Ya	Ya	login local	IZIN_REMOTE (VLAN 30 saja)
MLS1	Ya	Ya	Ya	login local	Tidak ada
SW1	Ya	Ya	Ya	login local	Tidak ada
SW2	Ya	Ya	Ya	login local	Tidak ada
SW3	Ya	Ya	Ya	login local	Tidak ada
ISP-Telkom	Di luar cakupan (bukan perangkat warnet)	Di luar cakupan	Di luar cakupan	login (default, tanpa password)	Di luar cakupan

Kondisi keamanan sudah meningkat signifikan dibanding revisi sebelumnya: seluruh perangkat internal (R1, MLS1, SW1–SW3) kini menggunakan service password-encryption, enable secret (hash), serta username dengan secret (hash) — tidak ada lagi password tersimpan sebagai plaintext ("password 0"). SSH versi 2 juga sudah konsisten aktif di semua switch, termasuk SW3 yang sebelumnya belum eksplisit. Telnet tetap dinonaktifkan di semua perangkat internal (transport input ssh).

Catatan desain: pada ISP-Telkom, baris "line vty 0 4 / login" sengaja dibiarkan tanpa password, sehingga akses Telnet/SSH ke perangkat tersebut otomatis ditolak ("% Login disabled - no password set"). Efeknya sama seperti menutup total akses remote — sesuai dengan tujuan bahwa ISP-Telkom tidak dikelola dari sisi warnet. Jika suatu saat perangkat ini perlu diakses untuk keperluan simulasi/troubleshooting, baru perlu ditambahkan password atau username tersendiri.

Lampiran: Running-Configuration Mentah

Bagian ini memuat salinan mentah hasil "show running-config" dari setiap perangkat sebagaimana diberikan, untuk keperluan referensi dan audit.

A.1 SW1

```
hostname SW1
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
ip ssh version 2
no ip domain-lookup
ip domain-name warnet.net
!
username admin secret 5 $1$mERr$J0cL4LHPKknNEQfY/3ZGI1
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/5
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/7
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/8
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/9
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/10
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/11
```

```
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 999
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport access vlan 999
```

```

switchport mode access
shutdown
!
interface GigabitEthernet0/2
switchport access vlan 999
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan30
ip address 172.16.30.3 255.255.255.0
!
ip default-gateway 172.16.30.1
!
line con 0
password 7 08364D5C071C11
login
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh

```

A.2 SW2

```

hostname SW2
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
ip ssh version 2
no ip domain-lookup
ip domain-name warnet.net
!
username admin secret 5 $1$mERr$J0cL4LHPKknNEQfY/3ZGI1
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/2
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/3
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/4
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/5
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/6
switchport access vlan 20
switchport mode access
!

```

```
interface FastEthernet0/7
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/8
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/9
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/10
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/11
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/12
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/13
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/14
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/15
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/16
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/17
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/18
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/19
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/20
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/21
  switchport access vlan 999
  switchport mode access
```

```

shutdown
!
interface FastEthernet0/22
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/24
switchport access vlan 999
switchport mode access
shutdown
!
interface GigabitEthernet0/1
switchport access vlan 999
switchport mode access
shutdown
!
interface GigabitEthernet0/2
switchport access vlan 999
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan30
ip address 172.16.30.4 255.255.255.0
!
ip default-gateway 172.16.30.1
!
line con 0
password 7 08364D5C071C11
login
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh

```

A.3 SW3

```

hostname SW3
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
ip ssh version 2
no ip domain-lookup
ip domain-name warnet.net
!
username admin secret 5 $1$mERr$J0cL4LHPKknNEQfY/3ZGI1
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/2

```

```
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/3
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/4
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/5
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
```

```
interface FastEthernet0/17
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/18
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/19
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/20
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/21
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/22
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/23
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface FastEthernet0/24
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface GigabitEthernet0/1
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface GigabitEthernet0/2
  switchport access vlan 999
  switchport mode access
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan30
  ip address 172.16.30.5 255.255.255.0
!
ip default-gateway 172.16.30.1
!
line con 0
  password 7 08364D5C071C11
  login
!
line vty 0 4
  login local
  transport input ssh
line vty 5 15
  login local
  transport input ssh
```

A.4 MLS1

```
hostname MLS1
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
no profinet
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
no ip cef
ip routing
!
no ipv6 cef
!
username admin secret 5 $1$mERr$J0cL4LHPKknNEQfY/3ZGI1
!
ip ssh version 2
no ip domain-lookup
ip domain-name warnet.net
!
spanning-tree mode pvst
!
interface GigabitEthernet1/0/1
 switchport mode trunk
!
interface GigabitEthernet1/0/2
 switchport mode trunk
!
interface GigabitEthernet1/0/3
 switchport mode trunk
!
interface GigabitEthernet1/0/24
 switchport mode trunk
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan30
 mac-address 0060.2fce.e901
 ip address 172.16.30.2 255.255.255.0
!
ip default-gateway 172.16.30.1
ip classless
!
ip flow-export version 9
!
line con 0
 password 7 08364D5C071C11
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
```

A.5 R1

```
hostname R1
!
```

```

version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
ip dhcp excluded-address 172.16.10.1 172.16.10.10
ip dhcp excluded-address 172.16.20.1 172.16.20.10
ip dhcp excluded-address 172.16.30.1 172.16.30.10
!
ip dhcp pool REGULER_POOL
network 172.16.10.0 255.255.255.0
default-router 172.16.10.1
dns-server 8.8.8.8
!
ip dhcp pool VIP_POOL
network 172.16.20.0 255.255.255.0
default-router 172.16.20.1
dns-server 8.8.8.8
!
ip dhcp pool MANAJEMEN_POOL
network 172.16.30.0 255.255.255.0
default-router 172.16.30.1
dns-server 8.8.8.8
!
ip cef
no ipv6 cef
!
username admin secret 5 $1$mERr$J0cL4LHPKknNEQfY/3ZGI1
license udi pid CISC02911/K9 sn FTX15240045-
!
ip ssh version 2
no ip domain-lookup
ip domain-name warnet.net
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 172.16.10.1 255.255.255.0
ip access-group REGULER in
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 172.16.20.1 255.255.255.0
ip access-group VIP in
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 172.16.30.1 255.255.255.0
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
!
interface Serial0/0/0
ip address 202.152.100.1 255.255.255.252
!
interface Serial0/0/1

```

```

no ip address
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ip nat inside source list NAT_ACL interface Serial0/0/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 202.152.100.2
!
ip flow-export version 9
!
ip access-list extended REGULER
deny ip 172.16.10.0 0.0.0.255 172.16.30.0 0.0.0.255
permit ip any any
ip access-list extended VIP
deny ip 172.16.20.0 0.0.0.255 172.16.30.0 0.0.0.255
permit ip any any
ip access-list standard NAT_ACL
permit 172.16.10.0 0.0.0.255
permit 172.16.20.0 0.0.0.255
permit 172.16.30.0 0.0.0.255
ip access-list standard IZIN_REMOTE
permit 172.16.30.0 0.0.0.255
!
line con 0
password 7 08364D5C071C11
login
!
line aux 0
!
line vty 0 4
access-class IZIN_REMOTE in
login local
transport input ssh

```

A.6 ISP-Telkom

```

hostname ISP-Telkom
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
no ip cef
no ipv6 cef
no ip domain-lookup
!
license udi pid CISCO2911/K9 sn FTX1524XWL4-
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0
no ip address
shutdown
!
interface GigabitEthernet0/1
ip address 8.8.8.1 255.255.255.0
!
interface GigabitEthernet0/2
no ip address
shutdown
!
interface Serial0/0/0
ip address 202.152.100.2 255.255.255.252
clock rate 64000

```

```
!  
interface Serial0/0/1  
  no ip address  
  clock rate 2000000  
  shutdown  
!  
ip classless  
ip route 172.16.10.0 255.255.255.0 202.152.100.1  
ip route 172.16.20.0 255.255.255.0 202.152.100.1  
ip route 172.16.30.0 255.255.255.0 202.152.100.1  
!  
line con 0  
!  
line vty 0 4  
  login
```